

Introduction to Probability and Applied Statistics

Week 3: Probability and data-generating processes

Melissa Sands

Department of Government, Cornell University · Fall 2026

Roadmap

Probability

Random variables and distributions

Data-generating processes

Wrapping up

Paul the octopus



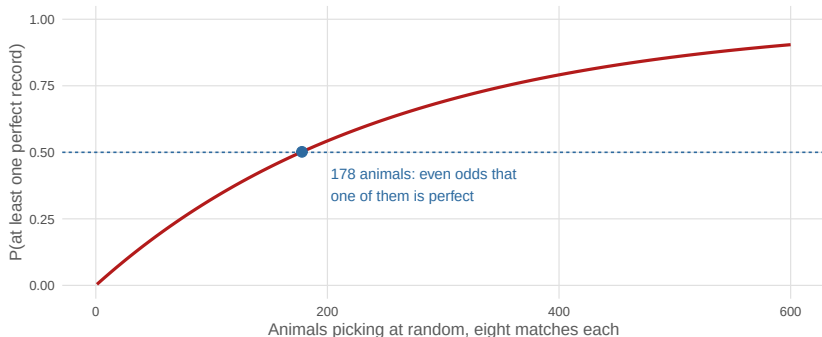
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Pattern or chance?

Round	Fixture	Paul's pick	Correct
Group	Germany–Australia	Germany	✓
Group	Germany–Serbia	Serbia	✓
Group	Ghana–Germany	Germany	✓
Last 16	Germany–England	Germany	✓
Quarter	Argentina–Germany	Germany	✓
Semi	Germany–Spain	Spain	✓
Third place	Uruguay–Germany	Germany	✓
Final	Netherlands–Spain	Spain	✓

Paul, an octopus at an Oberhausen aquarium, picked the winner of all eight matches put to him at the 2010 World Cup. Guessing at random gets all eight right with probability $2^{-8} = 1/256$.

A perfect record from nothing



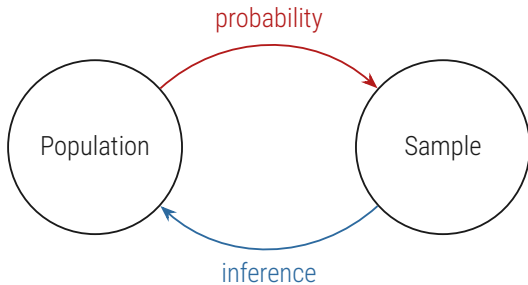
Leon the porcupine, Petty the pygmy hippo and Anton the tamarin missed every German group game; Mani the parakeet picked the Netherlands for the final.

The question this week answers

THE PROBLEM

Given a process that produces data, what outcomes should we expect, and how often?

Probability and inference



- **Inference:** learning about the population from a set of data.
- **Probability:** formalize the uncertainty about how our data came to be.

Sample space, outcome, event

DEFINITIONS

The **sample space** Ω is the set of all outcomes a process can produce. An **outcome** $\omega \in \Omega$ is one realization. An **event** $A \subseteq \Omega$ is any subset of the sample space.

One roll of a six-sided die: $\Omega = \{1, 2, 3, 4, 5, 6\}$, and $A = \{2, 4, 6\}$ is the event "the roll is even".

Two flips of a coin: $\Omega = \{HH, HT, TH, TT\}$, and $A = \{HT, TH\}$ is the event "exactly one head".

The probability axioms

KOLMOGOROV

For events in a sample space Ω :

1. $\mathbb{P}(A) \geq 0$ for every event A ;
2. $\mathbb{P}(\Omega) = 1$;
3. if $A_1, A_2, \dots$ are disjoint, $\mathbb{P}(\bigcup_i A_i) = \sum_i \mathbb{P}(A_i)$.

The third is **additivity**. **Disjoint** means no two of the events can happen at once, and for events like that the probability of getting one of them is the sum of the separate probabilities.

One die roll cannot be both a 1 and a 2, so
$$\mathbb{P}(\{1, 2\}) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}.$$



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The complement

$$\mathbb{P}(\Omega) = \mathbb{P}(A) + \mathbb{P}(A^c) = 1 \quad \Rightarrow \quad \mathbb{P}(A^c) = 1 - \mathbb{P}(A)$$

$\Omega = \{\text{Blue, Red, Green, Yellow}\}$ with $\mathbb{P}(\text{Blue}) = 0.25$ gives $\mathbb{P}(\text{not Blue}) = 0.75$, without knowing any of the other three individually.

Union and intersection

For events A and B , $A \cup B$ is “ A or B ” and $A \cap B$ is “ A and B ”. Both are symmetric: $\mathbb{P}(A \cap B) = \mathbb{P}(B \cap A)$.

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

One roll of a die, A = even, B = less than 4:

$$\mathbb{P}(A \cup B) = \frac{3}{6} + \frac{3}{6} - \frac{1}{6} = \frac{5}{6}$$

The subtracted term is the roll of 2, which is in both events and would otherwise be counted twice.

Two variables at once

IN THE SAMPLE (WHAT WE COMPUTED)

	Political interest			Total
	Not	Some	Very	
Democrat	0.03	0.16	0.27	0.46
Republican	0.02	0.09	0.28	0.39
Other	0.02	0.05	0.08	0.15
Total	0.06	0.30	0.64	1.00

Relative frequencies: counts divided by the 8,000 rows.

IN THE POPULATION (WHAT IT ESTIMATES)

	Political interest			Total
	Not	Some	Very	
Democrat	p_{11}	p_{12}	p_{13}	$p_{1\cdot}$
Republican	p_{21}	p_{22}	p_{23}	$p_{2\cdot}$
Other	p_{31}	p_{32}	p_{33}	$p_{3\cdot}$
Total	$p_{\cdot 1}$	$p_{\cdot 2}$	$p_{\cdot 3}$	1

The **joint distribution** of party and interest: p_{jk} is the probability that a person drawn at random falls in row j , column k .

Joint probability

JOINT PROBABILITY

Cross-classify a population by two characteristics. The probability of landing in a given cell is a **joint probability**, $\mathbb{P}(A \cap B)$, and the whole table of them is the **joint distribution**.

Voting and residence among U.S. citizens of voting age, 2024:

	In the U.S.	Abroad	Total
Voted	.630	.001	.631
Did not vote	.361	.008	.369
Total	.991	.009	1.000

Marginal probability

$$\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c)$$

The **marginal** distribution of one variable is the joint distribution summed over the other. It is the row and column totals, and it is what you would have had if you had never measured the second variable at all.

$$\mathbb{P}(\text{voted}) = .630 + .001 = .631$$

Conditional probability

CONDITIONAL PROBABILITY

For events A and B with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Conditioning restricts the sample space to B and renormalizes so the restricted space again has probability one.

Conditional probability, read off the table

	In the U.S.	Abroad	Total
Voted	.630	.001	.631
Did not vote	.361	.008	.369
Total	.991	.009	1.000

$$\mathbb{P}(\text{voted} \mid \text{abroad}) = \frac{.001}{.009} = .11 \quad \mathbb{P}(\text{abroad} \mid \text{voted}) = \frac{.001}{.631} = .002$$

Independence

INDEPENDENCE

X and Y are **independent**, written $X \perp\!\!\!\perp Y$, when

$$\mathbb{P}(A \mid B) = \mathbb{P}(A) \iff \mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B)$$

for all events A in X and B in Y .

Knowing B does not change $\mathbb{P}(A)$. In the table above $\mathbb{P}(\text{voted}) = .631$ but $\mathbb{P}(\text{voted} \mid \text{abroad}) = .11$, so voting and residence are not independent.

Example: U.S. Air Force missions, WWII

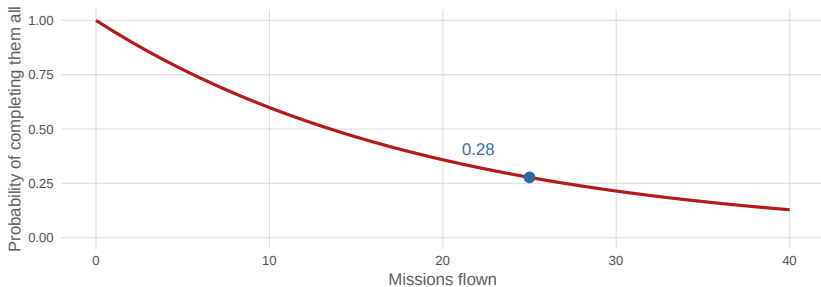


A tour of duty in the Eighth Air Force was twenty-five missions. The crew of the *Memphis Belle* finished theirs on May 19, 1943, among the first to do so, and was flown home for a thirty-one-city war-bond tour.

Compounding independent probabilities

Each mission carries a five percent chance of being lost, independent across missions:

$$\mathbb{P}(\text{complete all 25}) = .95^{25} = 0.28$$



When order matters

The joint probability always factors through the conditional:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B \mid A)$$

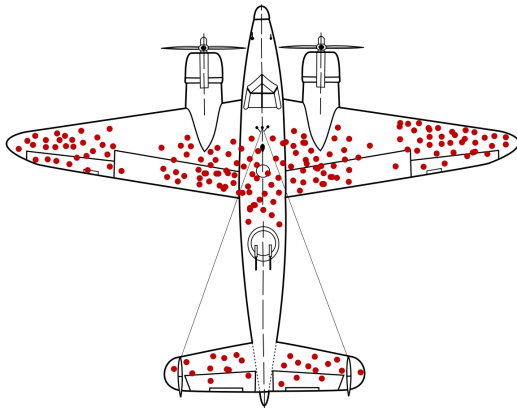
When A and B are independent, $\mathbb{P}(B \mid A) = \mathbb{P}(B)$ and this reduces to $\mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B)$.

Drawing two marbles without replacement from an urn holding four marbles, one of each of four colors,

$$\mathbb{P}(\text{Red then Yellow}) = \frac{1}{4} \cdot \frac{1}{3} = \frac{1}{12},$$

because the first draw changes the sample space for the second.

Conditioning on survival



Hypothetical damage on aircraft that returned from bombing missions, WWII.

McGeddon / Wikimedia Commons, CC BY-SA 4.0

What the holes are evidence of

Where on the planes should you add extra armor?

The data are $\mathbb{P}(\text{hit at } \ell \mid \text{returned})$ for a location ℓ on the plane; the question is $\mathbb{P}(\text{returned} \mid \text{hit at } \ell)$.

$$\mathbb{P}(A \mid B) \neq \mathbb{P}(B \mid A)$$

Engines carry almost no marks, and the reason is not that engines are rarely hit.

The answer: armor the parts with no holes in them.

Bayes' rule

Start from the joint probability, and factor it two ways.

$$\mathbb{P}(A \cap B)$$

A single cell of the crosstab.

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B) \mathbb{P}(B)$$

The definition of $\mathbb{P}(A \mid B)$, multiplied through by $\mathbb{P}(B)$.

$$\mathbb{P}(B \cap A) = \mathbb{P}(B \mid A) \mathbb{P}(A)$$

The same move with the roles swapped.

$$\mathbb{P}(A \mid B) \mathbb{P}(B) = \mathbb{P}(B \mid A) \mathbb{P}(A)$$

$A \cap B$ and $B \cap A$ are the same event, so the two factorizations are equal.

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A) \mathbb{P}(A)}{\mathbb{P}(B)}$$

Divide through by $\mathbb{P}(B)$, which the definition already required to be positive.

Expanding the denominator

$\mathbb{P}(B)$ is rarely handed to you. Expand it over A and A^c .

$$\mathbb{P}(B) = \mathbb{P}(B \cap A) + \mathbb{P}(B \cap A^c)$$

The marginal formula, taken over the partition $\{A, A^c\}$.

$$\mathbb{P}(B) = \mathbb{P}(B | A) \mathbb{P}(A) + \mathbb{P}(B | A^c) \mathbb{P}(A^c)$$

Each piece factored as before.

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(B | A) \mathbb{P}(A)}{\mathbb{P}(B | A) \mathbb{P}(A) + \mathbb{P}(B | A^c) \mathbb{P}(A^c)}$$

Substituted into the denominator. Every term on the right is $\mathbb{P}(A)$, $\mathbb{P}(A^c)$, or a conditional with B on the left. None of them requires $\mathbb{P}(B)$.

The denominator is every way B could have happened. The numerator is only those in which A happened too. Bayes' rule returns a share, so the answer depends on $\mathbb{P}(A)$ as much as on $\mathbb{P}(B | A)$.

Wald, worked

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A) \mathbb{P}(A)}{\mathbb{P}(B \mid A) \mathbb{P}(A) + \mathbb{P}(B \mid A^c) \mathbb{P}(A^c)}$$

Suppose hits are equally likely on the engine and on the fuselage, and that a plane hit in the engine returns with probability **.10** while a plane hit in the fuselage returns with probability **.90**.

$$\mathbb{P}(\text{engine} \mid \text{returned}) = \frac{.10 \times .50}{.10 \times .50 + .90 \times .50} = \frac{.05}{.50} = .10$$

One returning plane in ten was hit in the engine, though one in two of all planes hit was. The missing holes are the planes that did not come back.



Konrad Jacobs / MFO, CC BY-SA 2.0 DE

Screening for spies

Of 10,000 employees holding security clearances, ten are spies.

Set the polygraph to catch 80 of every 100 liars. At that setting it clears 84 of every 100 loyal employees.

One employee fails. The question is $\mathbb{P}(S \mid F)$, with S the event that they are a spy and F the event that they failed.



Security screening at the Clinton Engineer Works, Oak Ridge, c. 1945.

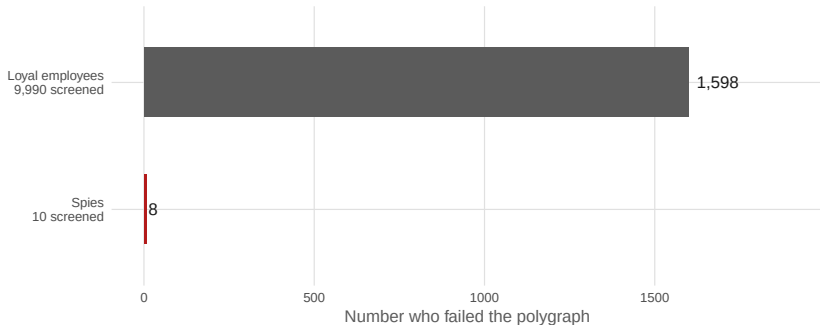
Screening, worked

$$\mathbb{P}(S \mid F) = \frac{\mathbb{P}(F \mid S) \mathbb{P}(S)}{\mathbb{P}(F \mid S) \mathbb{P}(S) + \mathbb{P}(F \mid S^c) \mathbb{P}(S^c)} = \frac{.80 \times .001}{.80 \times .001 + .16 \times .999}$$

$$\mathbb{P}(S \mid F) = \frac{.0008}{.0008 + .1598} = 0.005$$

One failed test in two hundred belongs to a spy.

Where the positives come from



Eight of the ten spies fail the test. So do 1,598 of the 9,990 loyal employees.

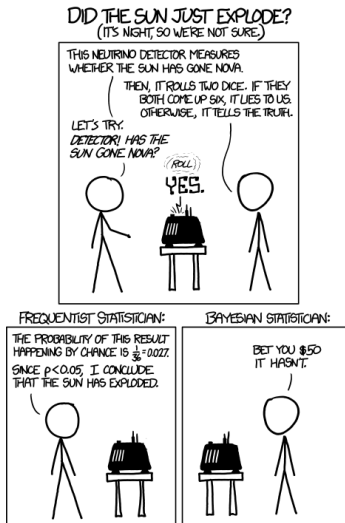
The bars are the two terms of the denominator, counted in people rather than written as probabilities: $10,000 \times .0008 = 8$ and $10,000 \times .1598 = 1,598$. Together they are everyone who fails, 1,606 of them. Only the carnelian bar, the eight spies, is also the numerator, and that ratio is the .005.

Prior, likelihood, posterior

$$\underbrace{\mathbb{P}(A | B)}_{\text{posterior}} \propto \underbrace{\mathbb{P}(B | A)}_{\text{likelihood}} \times \underbrace{\mathbb{P}(A)}_{\text{prior}}$$

$\mathbb{P}(A)$ is what you believed before seeing B ; $\mathbb{P}(A | B)$ is what you believe after.

Frequentists and Bayesians



Randall Munroe, xkcd.com/1132, CC BY-NC 2.5

Roadmap

Probability

Random variables and distributions

Data-generating processes

Wrapping up

Where we're at; where we're going

Feature	Sample	Population
Distribution	$\hat{F}(x) = \frac{1}{n} \sum_i \mathbf{1}\{X_i \leq x\}$	$F(x)$
Location	$\bar{X}$	$\mathbb{E}[X]$
Dispersion	s^2	$\mathbb{V}[X]$
Quantile	$\hat{Q}_p$	$Q_p(X)$

Random variables

RANDOM VARIABLE

A **random variable** is a function that assigns a number to each outcome in the sample space.

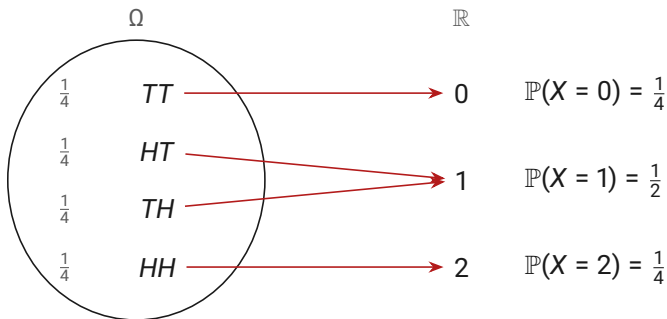
X denotes the variable, x a value it takes. Two coin flips with X = number of heads maps $TT \mapsto 0, HT \mapsto 1, TH \mapsto 1, HH \mapsto 2$.

The outcome HT is not itself a random variable: it is not a number.

A random variable is **discrete** when its range is countable – battle deaths in a civil war, seats won. It is **continuous** when $\mathbb{P}(X = x) = 0$ for every x , so probability is a property of intervals, not of points.

GDP per capita and a candidate's vote share are finely enough divided that we model them as continuous.

From outcomes to numbers



X = number of heads in two coin flips. The probability of a value is the total probability of the outcomes that map to it.

Where the randomness comes from

Draw one person at random from the U.S. citizen voting-age population, and let $X = 1$ if they voted in 2024, $X = 0$ otherwise.

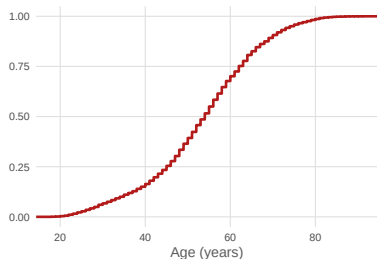
$$\mathbb{P}(X = 1) = .631$$

The probability is the share of the population holding that value. The randomness is in the draw, not in the person.

Let Y be the same person's age: $\mathbb{P}(Y > 65)$ is the share of that population over 65.

From the eCDF to the CDF

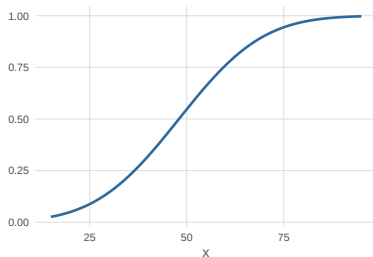
LAST WEEK



$$\hat{F}(x) = \frac{1}{n} \sum_i \mathbf{1}\{X_i \leq x\}$$

The share of the sample at or below x .

THIS WEEK



$$F(x) = \mathbb{P}(X \leq x)$$

The probability that the variable is at or below x .

The cumulative distribution function

CDF

For any random variable X , $F(x) = \mathbb{P}(X \leq x)$.

Three properties, all forced by the axioms:

- $F(x) \rightarrow 0$ as $x \rightarrow -\infty$ and $F(x) \rightarrow 1$ as $x \rightarrow +\infty$.
- F is weakly increasing: if $x_2 > x_1$ then $F(x_2) \geq F(x_1)$.
- $\mathbb{P}(X > x) = 1 - F(x)$, and $\mathbb{P}(a < X \leq b) = F(b) - F(a)$.

The CDF is defined for discrete and continuous variables alike.

From the CDF to the PMF

x	$f(x)$	$F(x)$
1	1/6	1/6
2	1/6	2/6
3	1/6	3/6
4	1/6	4/6
5	1/6	5/6
6	1/6	6/6

$$f(x_i) = F(x_i) - F(x_{i-1})$$

$$f(3) = \frac{3}{6} - \frac{2}{6} = \frac{1}{6}$$

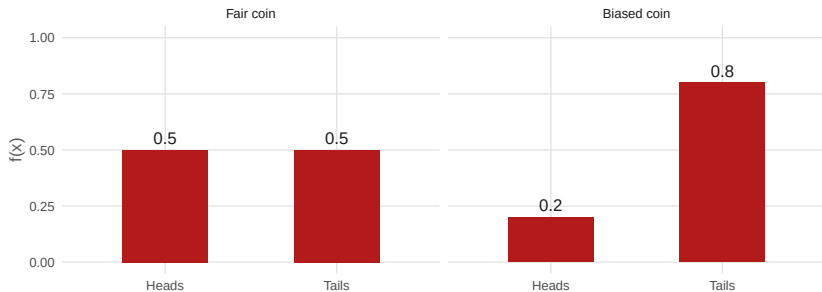
The probability mass function

PMF

For a discrete random variable, $f(x) = \mathbb{P}(X = x)$, with $0 \leq f(x) \leq 1$, $\sum_x f(x) = 1$, and $\mathbb{P}(X \in A) = \sum_{a \in A} f(a)$.

A function is a PMF if and only if it satisfies all three.

A theoretical variable



Note: Last week we saw empirical distributions, i.e., counts of things that happened. Here the distributions are **theoretical**, i.e., statements about what *could* happen.

Bernoulli Distribution

BERNOULLI

$X \sim \text{Bernoulli}(p)$ takes the value 1 with probability p and 0 with probability $1 - p$:

$$f(x; p) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}.$$

Anything with two outcomes: heads or tails, voted or did not, democracy or not. Week 2 showed that the mean of a 0/1 column is the share of ones; p is the population version of that share, and it is the only parameter the distribution has.

The outcome coded 1 is a **success** and the outcome coded 0 a **failure**; one run of the experiment is a **trial**. The labels are a convention: a success is whichever outcome the question counts, not a “good” outcome.

Counting sequences

Heads is the success. Three trials, two outcomes each: $2^3 = 8$ sequences.

Successes	Sequences			How many
0	TTT			1
1	HTT	THT	TTH	3
2	HHT	HTH	THH	3
3	HHH			1

Exactly two heads is not a sequence but an event: the set $\{HHT, HTH, THH\}$, three of the eight. That count is the **binomial coefficient**:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} \qquad \binom{3}{2} = \frac{3!}{2!1!} = \frac{6}{2} = 3$$

The count does not involve p : it is combinatorics, not probability. The three sequences are equally likely, so exactly two successes is three times the probability of any one of them.

Fifteen flips

Two sequences of fifteen flips of a fair coin, one ordered and one not:

O_1	H	H	H	H	H	H	H	H	T	T	T	T	T	T	T
O_2	H	T	T	H	T	H	T	H	H	H	T	T	H	H	T

Each has probability $2^{-15} = 1/32,768$. They are equally likely.

Binomial Distribution

BINOMIAL

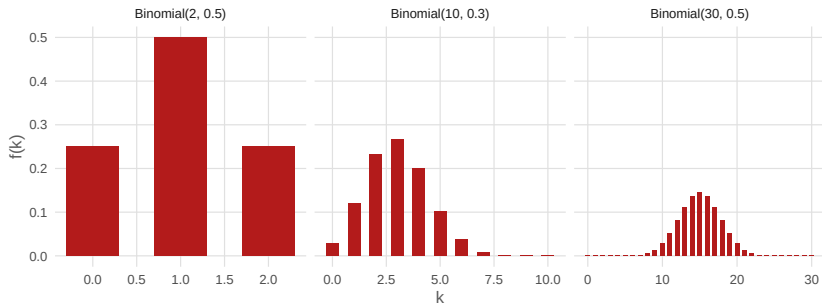
$X \sim \text{Binomial}(n, p)$ counts successes in n independent Bernoulli trials each with probability p :

$$f(k; n, p) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

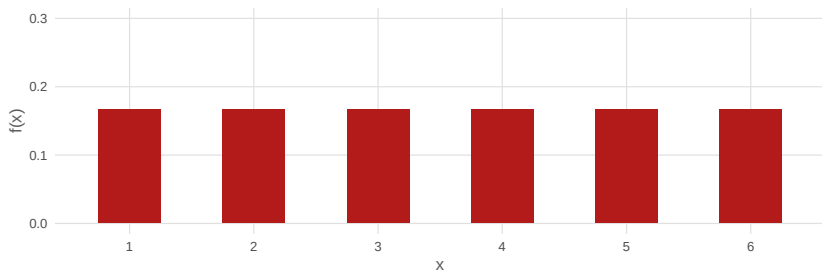
k is the number of successes. The coefficient counts the sequences; the rest is the probability of any one of them.

Independence across trials is an assumption (often doesn't hold in the real world, outside of coin flips).

Binomial Distribution



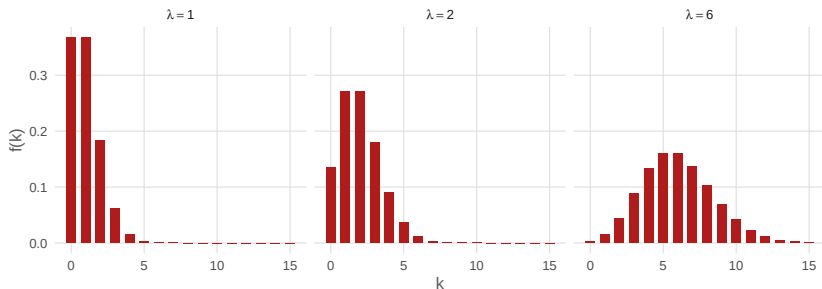
Discrete Uniform Distribution



$X \sim \mathcal{U}\{a, b\}$ with $f(x) = 1/n$ for each of the n values in the support.

With unequal probabilities attached to k named categories it is the **categorical** distribution, of which Bernoulli is the case $k = 2$.

Poisson Distribution



$X \sim \text{Poisson}(\lambda)$ with $f(k) = \lambda^k e^{-\lambda} / k!$ for $k \geq 0$: counts of events in a fixed window at an average rate λ .

Examples: terror attacks in a year, coups in a decade, protests in a city-month, 311 calls in a precinct-day.

Continuous random variables

For a continuous random variable the probability of any single value is zero. Probability is a property of intervals instead:

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f(x) dx = F(b) - F(a)$$

The **probability density function** $f(x)$ is the derivative of the CDF $F(x)$, and the limit of a histogram as the bins narrow.

A valid PDF has $f(x) \geq 0$ everywhere and encloses a total area of 1: $\int_{-\infty}^{\infty} f(x) dx = 1$. This is the continuous counterpart of $\sum_x f(x) = 1$ for a PMF.

A note on notation. Capital $F(x)$ is always the CDF: the probability that X is at most x . Lowercase $f(x)$ is the PMF for a discrete X , the PDF for a continuous one.

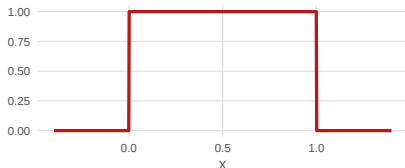
Continuous Uniform Distribution

UNIFORM

$X \sim \mathcal{U}(a, b)$ is a completely random number between a and b : the density is *constant* over the interval.

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a < x < b, \\ 0 & \text{otherwise.} \end{cases}$$

PDF



The area under the density is a rectangle, width $b - a$, height $1/(b - a)$. **Probability is proportional to length**: a subinterval of length ℓ has probability $\ell/(b - a)$.

The **standard uniform** $\mathcal{U}(0, 1)$ is the one we use most: $f(x) = 1$ on the unit interval.

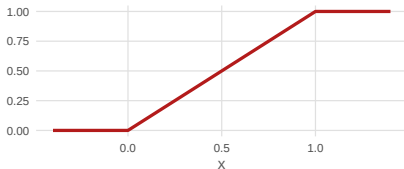
The uniform CDF

UNIFORM CDF

Integrating the constant density gives the CDF of $X \sim \mathcal{U}(a, b)$:

$$F(x) = \begin{cases} 0 & \text{if } x \leq a, \\ \frac{x-a}{b-a} & \text{if } a < x < b, \\ 1 & \text{if } x \geq b. \end{cases}$$

CDF



$F(x)$ is the area under the density to the left of x . The density is constant, so that area grows at a constant rate: a straight line from 0 at a to 1 at b , slope $1/(b - a)$.

For the standard uniform, $F(x) = x$. A draw falls below 0.3 with probability 0.3.

Standard Normal Distribution

STANDARD NORMAL

A continuous random variable Z has the **standard normal distribution** if its PDF is

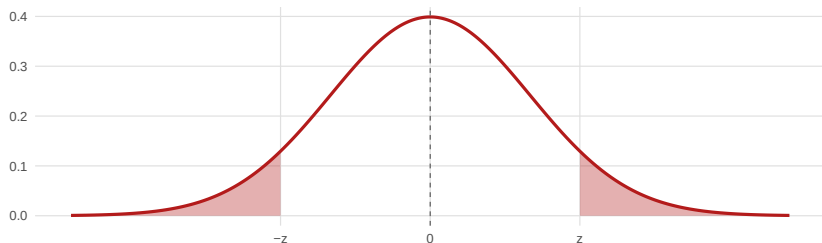
$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty.$$

We write $Z \sim \mathcal{N}(0, 1)$.

The $1/\sqrt{2\pi}$ in front is a **normalizing constant**. It is there for one reason: to make the total area under the PDF equal 1. It sets the height of the curve, not its shape.

The standard normal is used so often that its PDF and CDF have their own Greek letters: φ for the PDF, Φ for the CDF. Z is the conventional name for the variable itself.

Symmetry of the standard normal



- **Symmetry of the PDF:** $\varphi(z) = \varphi(-z)$, so the curve is a mirror image about 0 .
- **Symmetry of tail areas:** the area left of $-z$ equals the area right of z , which gives $\Phi(z) = 1 - \Phi(-z)$.
- **Symmetry of Z and $-Z$:** if $Z \sim \mathcal{N}(0, 1)$, then $-Z \sim \mathcal{N}(0, 1)$ as well.

From the standard normal to the Normal

NORMAL

If $Z \sim \mathcal{N}(0, 1)$, then

$$X = \mu + \sigma Z$$

has the **Normal distribution** with mean μ and variance σ^2 , for any real μ and σ^2 with $\sigma > 0$. We write $X \sim \mathcal{N}(\mu, \sigma^2)$.

Shifting by μ moves the center. Scaling by σ stretches the spread. Every normal is the standard normal moved and stretched.

Recall that s is the sample standard deviation, calculated from the data. Here σ is the population standard deviation. Note that we multiply Z by σ , not by the variance σ^2 ; else the units would be wrong and X would have variance σ^4 .

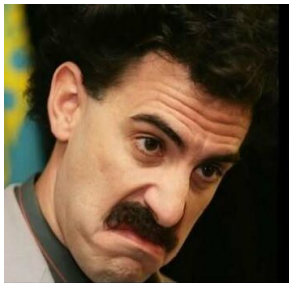
Standardization

Of course, if we can get from $\mathbf{Z}$ to $\mathbf{X}$, then we can get from $\mathbf{X}$ back to $\mathbf{Z}$. The process of getting a standard Normal from a non-standard Normal is called, appropriately enough, *standardization*.

For $\mathbf{X} \sim \mathcal{N}(\mu, \sigma^2)$, the *standardized version* of $\mathbf{X}$ is

$$\frac{\mathbf{X} - \mu}{\sigma} \sim \mathcal{N}(0, 1).$$

Standardization, the short way



$$\begin{aligned}
 F_X(x) &= \int_{-\infty}^x \mathcal{N}(z; \mu, \sigma^2) dz \\
 &= \int_{-\infty}^x \frac{1}{\sqrt{2\pi}\sigma} \cdot \exp\left[-\frac{1}{2}\left(\frac{z-\mu}{\sigma}\right)^2\right] dz \\
 &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^x \exp\left[-\left(\frac{z-\mu}{\sqrt{2}\sigma}\right)^2\right] dz \\
 &= \frac{1}{\sqrt{2\pi}\sigma} \int_{(-\infty-\mu)/(\sqrt{2}\sigma)}^{(x-\mu)/(\sqrt{2}\sigma)} \exp(-t^2) d(\sqrt{2}\sigma t + \mu) \\
 &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\frac{x-\mu}{\sqrt{2}\sigma}} \exp(-t^2) dt \\
 &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^s \exp(-t^2) dt + \frac{1}{\sqrt{\pi}} \int_0^{\frac{x-\mu}{\sqrt{2}\sigma}} \exp(-t^2) dt \\
 &= \frac{1}{\sqrt{\pi}} \int_0^{\infty} \exp(-t^2) dt + \frac{1}{\sqrt{\pi}} \int_0^{\frac{x-\mu}{\sqrt{2}\sigma}} \exp(-t^2) dt \\
 &= \frac{1}{2} \lim_{s \rightarrow \infty} \operatorname{erf}(s) + \frac{1}{2} \operatorname{erf}\left(\frac{x-\mu}{\sqrt{2}\sigma}\right) \\
 &= \frac{1}{2} \left[1 + \operatorname{erf}\left(\frac{x-\mu}{\sqrt{2}\sigma}\right) \right] \\
 &= \Phi\left(\frac{x-\mu}{\sigma}\right)
 \end{aligned}$$



$$\begin{aligned}
 F_X(x) &= P(X \leq x) \\
 &= P\left(Z \leq \frac{x-\mu}{\sigma}\right) \\
 &= \Phi\left(\frac{x-\mu}{\sigma}\right)
 \end{aligned}$$

Normal Distribution

NORMAL

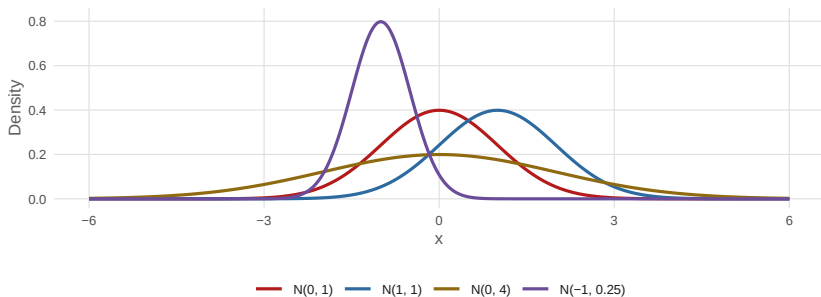
$X \sim \mathcal{N}(\mu, \sigma^2)$ has density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{x-\mu}{\sigma}\right)^2\right).$$

The case $\mu = 0, \sigma^2 = 1$ is the **standard normal**.

Recall that s^2 is the sample variance, calculated from the data. Here σ^2 is the population variance.

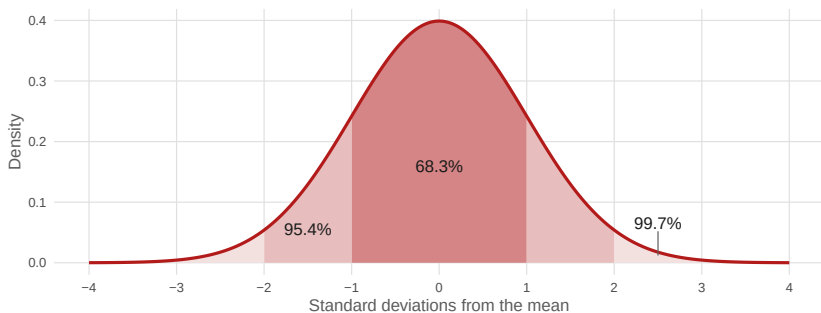
Normal Distribution



$\mathcal{N}(-1, 0.25)$ moves both: left of 0 and narrower.

Symmetric about μ , so the mean, the median and the mode coincide.

The standard normal, $\mathcal{N}(0, 1)$



You'll see this again, as the 68–95–99.7 rule. In a normally distributed population, ~68% of observations fall within one standard deviation of the mean, 95% within two, 99.7% within three.

Reading the normal in R

```
pnorm(1) - pnorm(-1)          # P(-1 <= X <= 1) for a standard normal
```

```
[1] 0.6826895
```

```
qnorm(0.975)                  # the value with 97.5% of the mass below it
```

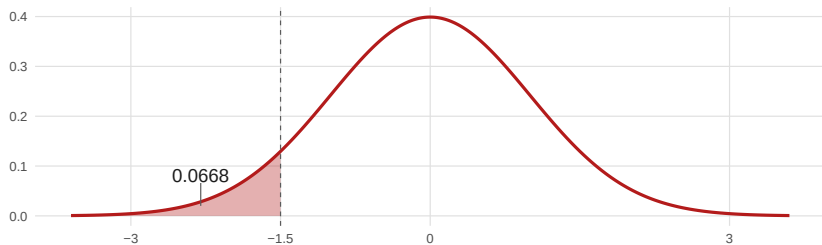
```
[1] 1.959964
```

`p` gives the CDF, `q` its inverse, `d` the density, `r` a random draw. Every distribution in R follows the same four-letter convention: `pnorm`, `dbinom`, `rpois`, `qunif`.

pnorm: the area to the left

```
pnorm(-1.5)
```

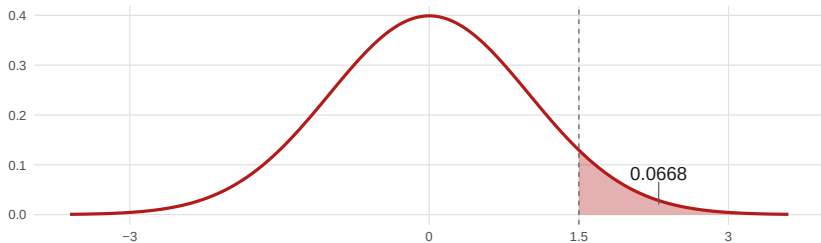
```
[1] 0.0668072
```



pnorm: the area to the right

```
pnorm(1.5, lower.tail = FALSE)
```

```
[1] 0.0668072
```

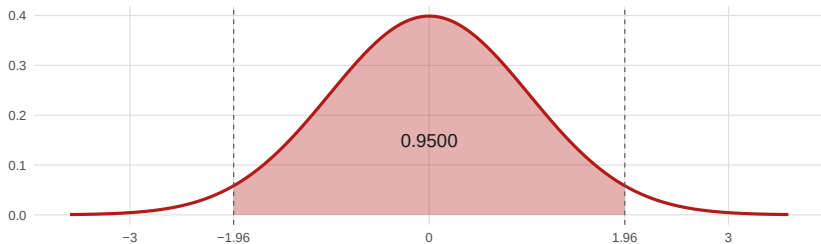


1 - pnorm(1.5) gives the same number.

pnorm: the area between

```
pnorm(1.96) - pnorm(-1.96)
```

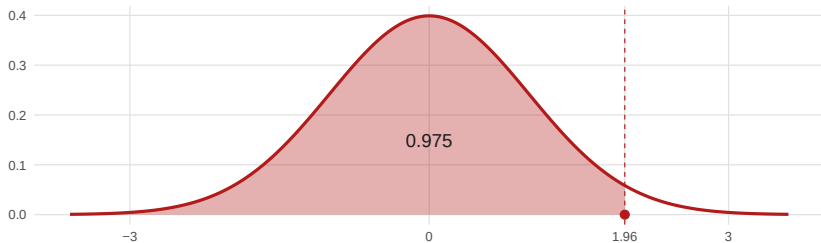
```
[1] 0.9500042
```



qnorm: from probability back to cutoff

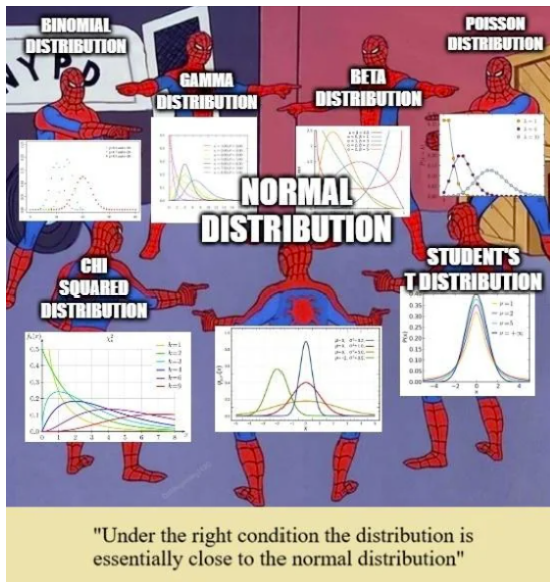
```
qnorm(0.975)
```

```
[1] 1.959964
```



`pnorm` takes the cutoff and returns the area. `qnorm` takes the area and returns the cutoff.

Where we're going



Roadmap

Probability

Random variables and distributions

Data-generating processes

Wrapping up

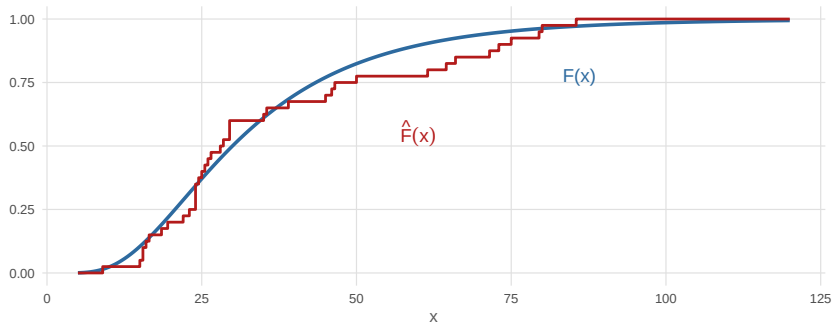
The data-generating process

DATA-GENERATING PROCESS

The **data-generating process** is the mechanism, in the world and in the study design, that produced the rows you have. Gailmard calls it the set of rules that produce data.

You never observe it. You observe one dataset it produced.

The process fixes F ; the sample gives you $\hat{F}$



Components of a process

Every part of a process is either fixed or the product of chance.

- **Deterministic**: typically 'fixed' or 'given'
- **Probabilistic**: product of a probabilistic process

These probabilistic components can themselves be of different types:

- **Systematic**: product of other variables, of which at least one is probabilistic
- **Stochastic**: the randomness that remains once the systematic part is fixed

A systematic component is a deterministic function of random inputs. It varies across datasets only because its inputs do.

A process for turnout

```
n    <- 5000                                # deterministic: the frame
age  <- sample(18:85, n, replace = TRUE)    # stochastic
ed   <- rbinom(n, 1, 0.32)                  # stochastic

p    <- 0.20 + 0.004 * (age - 18) + 0.15 * ed # systematic
V    <- rbinom(n, 1, p)                      # stochastic

sum(V)                                     # how many voted
```

```
[1] 1910
```

`sample()` draws from a discrete uniform, `rbinom(n, 1, .)` from a Bernoulli.

What each line is

Object	Component	Why
n , the 18+ rule	Deterministic	Fixed by the design, not drawn
age, ed	Stochastic	Drawn, independent of the rest
p	Systematic	A function of age and ed
v	Stochastic	A Bernoulli draw, given p

The last line of the process

```
reported <- ifelse(V == 1, 1, rbinom(n, 1, 0.12))
```

Twelve percent of non-voters say they voted. That line is not about the electorate; it is about the survey. It is part of the process all the same, and **reported**, not **V**, is what you are handed.

The DGP includes...

WHAT HAPPENS IN THE WORLD

People really do differ in age and education, and those differences really do bear on whether they vote.

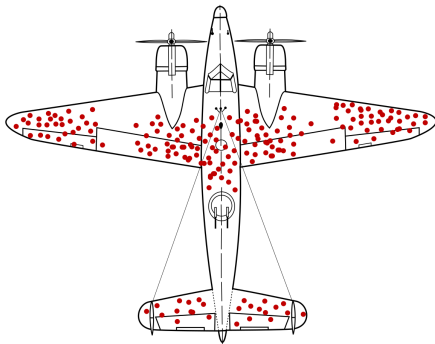
The object of inquiry.

WHAT HAPPENS IN THE STUDY

The sampling frame, the recruitment, the question wording, who picked up the phone, who told the truth.

Both are in the data-generating process. Only one of them is what you set out to learn about, and no amount of sample size separates them.

Where the holes are



The returning aircraft are a sample, and returning is part of the process that produced it. Something determined which rows you got to see.

McGeddon / Wikimedia Commons, CC BY-SA 4.0

Roadmap

Probability

Random variables and distributions

Data-generating processes

Wrapping up

Further reading

- Gailmard, *Statistical Modeling and Inference for Social Science*, ch. 3.
- Aronow and Miller, *Foundations of Agnostic Statistics*, ch. 1.
- Blitzstein and Hwang, *Introduction to Probability*:
<https://probabilitybook.net/>.
- Wald (1943), "A Method of Estimating Plane Vulnerability Based on Damage of Survivors."
- Clauset, Shalizi and Newman (2009), "Power-Law Distributions in Empirical Data."
- National Research Council (2003), *The Polygraph and Lie Detection*.

These slides draw on teaching materials by Talbot Andrews, Matt Blackwell, Joe Blitzstein, Daniel de Kadt, Tom Pepinsky and Teppei Yamamoto.